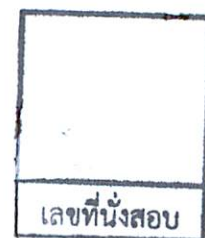


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Subject: 392153 Chemistry III

Date: 14 October 2015

Year: 2015

Section: 15 - 16

Time: 13.00 – 15.00

Name: _____ ID: _____ Calss: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks are **NOT** allowed.
4. Dictionaries are **NOT** permitted.
5. A calculator **NOT** permitted.
6. Electronic communication device are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page and periodic table), a total score of 65 marks.
8. Write all your answer on this examination sheet.

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Value: 65 marks

Suggested Time: 120 minutes

INSTRUCTIONS: Written Response

Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.

1. Match description on the left to the appropriate term on the right (6 marks)

Please note that: not all terms will be used

Answer	Description	Term
	holds molecules together	Ionic compound
	has a high melting point and conducts electricity when dissolved	Ionic bond
	never forms compounds	Covalent compound
	forms when pairs of electrons are shared by two atoms	Covalent bond
	forms when electrons are transferred between atoms	Metals
	is according to noble gas	Intermolecular force
		Attraction
		Inert gas
		Octet rule

2. Modified True/False (6 marks)

Indicate whether the sentence or statement is true or false.

If false, change the identified word or phrase to make the sentence or statement true.

True/False	State	Revise
	Particles formed from covalent bonding of atoms are called <u>ions</u>	
	Molecules are <u>neutral</u>	
	When an atom gains or loses electrons, the charged particle is called a <u>molecule</u>	
	An atom is chemically <u>unstable</u> when its outer energy level contains all the electrons it can hold	
	An element's <u>oxidation number</u> indicates how many electrons the element must gain, lose, or share to become stable.	

True/False	State	Revision
	A <u>chemical formula</u> is the force that holds atoms together in a compound.	

3. Giving short explanation (8 marks)

3.1 What makes a compound different from an element?

3.2 What makes an ionic compound different from a covalent compound?

3.3 What makes solutions different from other mixtures?

3.4 Order the following bonds according to polarity: HOH, OOH, ClOH, SOH, and FOH.

4. Give possible Lewis structures for XeO_3 , an explosive compound of xenon.

(Hint: 8 structures are needed to write) (10 marks)

Which Lewis structure or structures are most appropriate according to the formal charges?

5. Which of the following compounds would you expect to be soluble in SBr_2 (a bent molecule)? H_2O , SeBr_2 , SF_2 , NF_3 (5 marks)

6. For each of the following molecules, determine if the bonds are nonpolar covalent or polar covalent. Next, determine a dipole moment of the molecule. (10 marks)

Compound	molecular shape	bond type	dipole moment
SiO ₂	linear	_____	_____
NF ₃	trigonal pyramidal	_____	_____
SO ₂	bent	_____	_____
CH ₄	tetrahedral	_____	_____
SO ₃	trigonal planar	_____	_____

7. What type(s) of intermolecular forces exist between the following pairs? (10 marks)

(a) HBr and H₂S, (b) Cl₂ and CBr₄, (c) I₂ and NO₂, (d) NH₃ and C₆H₆

8. Calculate lattice energy of LiF(s) (10 marks)

Giving:

Process	Energy change (kJ)
$\text{Li}(s) \rightarrow \text{Li}(g)$	161
$\text{Li}(g) \rightarrow \text{Li}^+(g) + e^-$	520
$\frac{1}{2}\text{F}_2(g) \rightarrow \text{F}(g)$	77
$\text{F}(g) + e^- \rightarrow \text{F}^-(g)$	-328
$\text{Li}^+(g) + \text{F}^-(g) \rightarrow \text{LiF}(s)$	-1047
Overall: $\text{Li}(s) + \frac{1}{2}\text{F}_2(g) \rightarrow \text{LiF}(s)$	-617 kJ (per mole of LiF)

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45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.8
77 Ir	78 Pt	79 Au	80 Hg
109 Mt	110	111	112

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